Distributional Families in zzrctsim and the Boundaries of the Gaussian Machinery

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1 Purpose

`zzrctsim` generates longitudinal trial data. The conditional generator now accepts a `family` and `link`, so Gaussian, binomial, Poisson and negative binomial responses can be produced. Extending the generator does **not** extend the rest of the package: several downstream operations are defined only for a Gaussian response, and running them on a count or binary outcome would return plausible numbers that mean nothing.

This report records which families are supported, why the exponential family unifies less than it appears to, exactly which operations are Gaussian-only, and how the package refuses work it cannot do.

2 What the exponential family does and does not unify

The exponential family unifies *univariate marginal* distributions. It supplies no multivariate analogue: there is no canonical multivariate binomial or multivariate Poisson corresponding to the multivariate normal.

For a Gaussian response the marginal family and the dependence structure are the same object, the covariance matrix. For every other family they decouple, and a separate dependence mechanism must be chosen. That choice, and not the marginal law, is the design decision.

`zzrctsim` takes the conditional route. Random effects are drawn, `b_i ~ N(0, G)`, and the response is drawn conditionally, `y_ij ~ family(g^{-1}(eta_ij))` with `eta_ij = x_ij'beta + z_ij'b_i`. This is the GLMM construction used by `lme4` and `glmmTMB`.

Two consequences follow, and both belong to the user rather than to the package.

First, dependence between visits arises solely through the shared `b_i`. The reachable dependence is exactly what the random-effects structure can induce, and there is no marginal escape hatch: `dgm_marginal()` has no meaning outside the Gaussian case.

Second, under a non-identity link the marginal mean is not `g^{-1}(x'beta)`. The fixed effects are conditional on the random effects, so conditional and marginal treatment effects differ in magnitude. An `estimand()` used with a non-Gaussian family must state which of the two it names.

3 Supported families

```
s <- trial_schedule(treatment = 4, interval = 3)
arm <- factor(rep(c('placebo', 'active'), each = 300),
              levels = c('placebo', 'active'))
d <- runin_design(s, arm, reference = 'placebo')
ri <- function(t) matrix(1, length(t), 1)
b <- c(x_slope = 0.02, x_hinge = 0, x_trt_active = -0.05)

specs <- list(
  list(nm = 'gaussian(identity)', f = gaussian(),         s2 = 4,    ic = 20),
  list(nm = 'binomial(logit)',    f = binomial(),         s2 = NULL, ic = 0),
  list(nm = 'binomial(probit)',   f = binomial('probit'), s2 = NULL, ic = 0),
  list(nm = 'poisson(log)',       f = poisson(),          s2 = NULL, ic = 1)
)

set.seed(8)
out <- do.call(rbind, lapply(specs, function(sp) {
  g <- dgm_conditional(G = matrix(0.5), sigma2 = sp$s2,
                       z = ri, family = sp$f)
  o <- generate_outcomes(d, g, beta = b, intercept = sp$ic)
  w <- matrix(o$y, ncol = nrow(s), byrow = TRUE)
  data.frame(family = sp$nm, min = min(o$y), max = max(o$y),
```

```
             mean = mean(o$y), within_cor = cor(w[, 2], w[, 3]))
}))
knitr::kable(out, digits = 3, booktabs = TRUE,
             caption = 'Response support and within-subject
                        dependence by family.')
```

Table 1: Response support and within-subject dependence by family.

family	min	max	mean	within_cor
gaussian(identity)	11.8	27.31	20.067	0.162
binomial(logit)	0.0	1.00	0.481	0.092
binomial(probit)	0.0	1.00	0.487	0.196
poisson(log)	0.0	38.00	3.417	0.663

The within-subject correlation is induced by the shared random intercept in every case. It is not a free parameter: it is whatever the random-effects structure implies on the response scale, and it differs by family even with `G` held fixed.

Negative binomial responses require `theta`, the dispersion in the `size` parameterization of `rnbinom()`. Binomial responses accept `trials`, defaulting to `1` for a binary outcome.

4 Three scales, all retained

The GLMM path returns three columns rather than collapsing them, because for a non-identity link they are genuinely different quantities and a reader cannot reconstruct one from another without knowing the link.

```
set.seed(21)
gb <- dgm_conditional(G = matrix(0.5), z = ri, family = binomial())
ob <- generate_outcomes(d, gb, beta = b, intercept = 0)
knitr::kable(head(ob[, c('id', 'time', 'mu', 'eta', 'cmean', 'y')], 6),
             digits = 3, booktabs = TRUE,
             caption = 'mu is the fixed-effect linear predictor, eta
                        adds the random effects, cmean applies the
                        inverse link, y is the draw.')
```

Table 2: mu is the fixed-effect linear predictor, eta adds the random effects, cmean applies the inverse link, y is the draw.

id	time	mu	eta	cmean	y
1	0	0.00	0.561	0.637	1
1	3	0.06	0.621	0.650	1
1	6	0.12	0.681	0.664	0
1	9	0.18	0.741	0.677	0
1	12	0.24	0.801	0.690	1
2	0	0.00	0.369	0.591	0

id	time	mu	eta	cmean	y

The realized random effects are retained on a `ranef` attribute, so a replicate can be interrogated rather than only re-run.

5 What is Gaussian-only, and why

Four operations presuppose a Gaussian response. Each is now guarded.

5.1 Covariance of the response

`cov_at()` returns the covariance of the response. For a binomial or Poisson response the variance is determined by the mean, so it cannot be specified separately, and the function refuses.

```
cov_at(gb, s$time)
```

```
## Error in `cov_at.dgm_conditional()`:
## ! `cov_at()` is the covariance of the response and is defined only for the Gaussian famil
```

`linpred_cov()` is the defined alternative for every family: it returns `Z G Z'`, the covariance induced on the linear-predictor scale.

```
linpred_cov(gb, s$time)[1:3, 1:3]
```

```
##      [,1] [,2] [,3]
## [1,]  0.5  0.5  0.5
## [2,]  0.5  0.5  0.5
## [3,]  0.5  0.5  0.5
```

5.2 Conversion and the reachability certificate

`certify()`, `as_conditional()` and `as_marginal()` are results about `V = Z G Z' + sigma^2 I`. They are Gaussian statements and have no non-Gaussian analogue, because the marginal parameterization they convert to does not exist outside that case.

5.3 Reference-based post-discontinuation trajectories

`reference_based()` preserves each subject's additive residual `y - mu` while shifting the mean. On a count or binary scale that construction produces values outside the support: a probability shifted by a residual is not a probability, and a shifted count need not be an integer. The function refuses rather than returning impossible data.

```
mk <- dropout_mask(ob, target = 0.30)
reference_based(ob, mk, method = 'j2r')
```

```
## Error:
## ! `reference_based()` is defined only for a Gaussian response; this data set was generate
##   The construction preserves each subject's additive residual `y - mu`, which is not mean
```

5.4 Response-dependent missingness

The dropout hazard follows Diggle and Kenward (1994),

$$\operatorname{logit} P(\text{drop at } j) = \psi_0 + \psi_1(y_{j-1} - c) + \psi_2(y_j - c),$$

with the response centred at `c`. Two things are then scale-dependent: the default centring, the mean of `y`; and the interpretation of `psi1` as a change in log-odds per unit of `y`.

The guard is deliberately narrow. **MCAR is permitted for every family**, because its hazard never reads `y` at all. Only response-dependent missingness is restricted, and it is permitted for a non-Gaussian family as soon as `center` is supplied explicitly, since supplying it is the act of deliberation the guard exists to force.

```
guard <- function(label, expr) {
  r <- tryCatch({ expr; 'allowed' },
                error = function(e) 'blocked')
  data.frame(call = label, result = r)
}
knitr::kable(rbind(
  guard('binomial, MCAR',               dropout_mask(ob, target = 0.2)),
  guard('binomial, MAR, no center',     dropout_mask(ob, target = 0.2,
                                                     psi1 = 0.3)),
  guard('binomial, MAR, center given', dropout_mask(ob, target = 0.2,
                                                     psi1 = 0.3,
                                                     center = 0.5)),
  guard('binomial, reference_based',   reference_based(ob, mk, 'j2r'))
), booktabs = TRUE, row.names = FALSE,
   caption = 'Guards permit MCAR for any family and require an
             explicit centring for response-dependent mechanisms.')
```

Table 3: Guards permit MCAR for any family and require an explicit centring for response-dependent mechanisms.

call	result
binomial, MCAR	allowed
binomial, MAR, no center	blocked
binomial, MAR, center given	allowed
binomial, reference_based	blocked

6 What is family-agnostic

The guards are narrow because most of the package makes no distributional assumption at all. The design, missingness-application, driver and performance layers operate on estimates and indices, not on a response distribution:

Layer	Functions	Family-agnostic
Schedule	`trial_schedule`, `runin_design`	yes
Accrual	`accrue`, `close_out`, `realize_schedule`	yes
Mask application	`apply_mask`, `mask_from_data`	yes
Driver	`run_simulation`, `sim_streams`, `with_rng_state`	yes
Performance	`compute_performance`, `nsim_for_mcse`	yes
Power	`sim_power`, `power_curve`, `sample_size`	yes
Generation	`generate_outcomes`	family-aware
Covariance	`cov_at`, `certify`, `as_conditional`	Gaussian only
Mask generation	`dropout_mask`, `apply_dropout`	Gaussian, or explicit centring
Pattern mixture	`reference_based`	Gaussian only

This is why the endpoint families of the design document are described as *five generators behind one contract* rather than one generator with five settings. The reuse lives in the spine, which is built; the variation lives in the generator, which is not yet complete.

The spine carries a non-Gaussian study end to end today:

```
fit_glm <- function(z) {
  zz <- z[abs(z$time - max(z$time)) < 1e-8, ]
  m <- tryCatch(glm(y ~ arm, data = zz, family = binomial()),
                error = function(e) NULL)
  if (is.null(m)) return(null_fit())
  cf <- summary(m)$coefficients
  fit_result(cf[2, 'Estimate'], cf[2, 'Std. Error'], cf[2, 'Pr(>|z|)'])
}
res <- run_simulation(
  B = 300,
  generate = function() generate_outcomes(d, gb, beta = b, intercept = 0),
  analyse = list(glm = fit_glm),
  estimand = estimand('conditional log-OR at final visit', NA_real_),
  seed = 2L
)
perf <- compute_performance(res, theta = 0)
knitr::kable(perf[perf$measure %in% c('n_converged', 'emp_se',
                                      'mod_se', 'rejection'),
```

```
                 c('measure', 'estimate', 'mcse')],
            digits = 4, booktabs = TRUE, row.names = FALSE,
            caption = 'Morris Table 6 measures for a binary endpoint.')
```

Table 4: Morris Table 6 measures for a binary endpoint.

measure	estimate	mcse
n_converged	300.0000	NA
emp_se	0.1648	0.0067
mod_se	0.1652	0.0001
rejection	0.9100	0.0165

Note that `theta` is passed as `0` for a null check rather than as a marginal log-odds ratio. The estimand under a logit link is ambiguous between the conditional and marginal effect, and the package does not resolve that ambiguity on the user's behalf.

7 Boundaries and open work

Two of the five planned endpoint families fall outside the exponential family altogether and will need their own generators rather than another `family` argument.

Time-to-event. Censoring, hazards and risk sets are a different structure, not a different marginal law. Required by compendia 02, 10, 13 and 20.

Mixtures of alpha-stable laws. No closed-form density, and for stability index below two no finite variance. That last point reaches beyond the generator: without moments, bias, empirical standard error and mean squared error are undefined, so `compute_performance()` must dispatch on whether moments exist rather than return a finite-looking number. This is the one place where an endpoint family breaks the family-agnostic spine.

A copula route would allow arbitrary marginals with a specified correlation, but the requested correlation is not the achieved one for discrete responses, and the shortfall worsens as events become rarer. If that route is added, the achieved correlation must be computed and returned rather than assumed.

8 Reproducibility

```
## zzrctsim: development version
```

```
## R: R version 4.6.1 (2026-06-24)
```

Source: *`~/prj/sfw/17-zzrctsim/zzrctsim/analysis/report/report.Rmd`*